

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run	Yes
3	requirements.txt / dependency file	Yes
4	requirements_full.txt (for full ML training)	Yes
5	.env.example (API key template)	Yes
6	.gitignore	Yes
7	streamlit_app.py (cloud entry point)	Yes
8	app/main.py (local entry point)	Yes
9	config/settings.py (all configuration)	Yes
10	config/emotions.py (emotion definitions)	Yes
11	src/ (all 20+ Python modules)	Yes
12	tests/ (25+ unit tests)	Yes
13	pytest.ini (test configuration)	Yes
14	All 8 Phase documentation templates	Yes

Step 2: How to Run the Project

Visit the deployed Streamlit app:
<https://Harsh-maker007-Smartbridge-project1-streamlit-app.streamlit.app>

- Paste your Gemini API key in the input box
- Type your study problem
- Click "Detect Emotion"

Option B: Local Run (Demo Mode)

```
# 1. Clone the repository
git clone https://github.com/Harsh-maker007/Smartbridge_project1.git
cd Smartbridge_project1

# 2. Create and activate virtual environment
python -m venv venv
venv\Scripts\activate # Windows

# 3. Install lightweight dependencies
pip install -r requirements.txt

# 4. Set your Gemini API key
copy .env.example .env
# Edit .env: add GEMINI_API_KEY=your_key_here

# 5. Run the app
streamlit run streamlit_app.py
```

Option C: Full Local Run (BiLSTM + BERT Training)

```
# Install full ML dependencies
pip install -r requirements_full.txt

# Download and preprocess dataset
python scripts/download_data.py

# Train BiLSTM model (~10 min on GPU)
python scripts/train_bilstm.py

# Fine-tune BERT model (~30 min on GPU)
python scripts/train_bert.py

# Run full-mode app (uses trained weights)
streamlit run app/main.py
```

Step 3: Key Executable Files

File	Type	Purpose
`streamlit_app.py`	Entry Point	Cloud/demo mode launcher

File	Type	Purpose
`app/main.py`	Entry Point	Local full-mode launcher
`scripts/download_data.py`	Script	Download HuggingFace dataset
`scripts/train_bilstm.py`	Script	Train BiLSTM model
`scripts/train_bert.py`	Script	Fine-tune BERT model
`requirements.txt`	Config	Cloud dependencies
`requirements_full.txt`	Config	Full ML dependencies
`.env.example`	Template	API key configuration

Step 4: GitHub Repository

Field	Value
Repository URL	https://github.com/Harsh-maker007/Smartbridge_project1
Branch	main
Total Commits	7+
Total Files	71+
Language	Python 100%
Deployment	Streamlit Community Cloud (Free)